Diffraction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain spreading around edges and through openings.

SUCCESS CRITERIA - I can...

1. Compare opening size with wavelength.
2. Distinguish diffraction from refraction.
3. Use a wave picture to justify spreading.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A wave travels 9 m in 0.3 s at frequency 50 Hz. Find speed and wavelength.

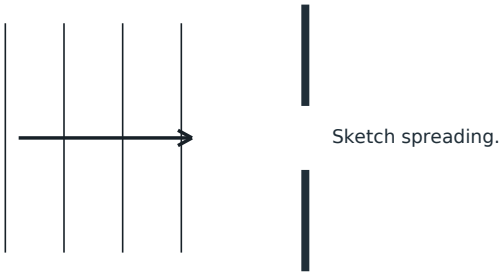
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\lambda = v/f$; compare opening width with λ. Spreading is pronounced when the opening width is comparable to wavelength. Label the opening and wavefronts.

Worked reasoning: A wave with wavelength 1 m meets openings 1 m and 10 m wide. Spreading relative to opening size is more noticeable for the 1 m opening.

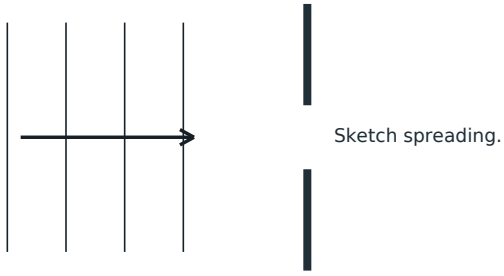
Your turn: Draw incoming parallel wavefronts and the emerging patterns for each opening.



Marzano check - circle: ① ② ③

03 Read and interpret the model

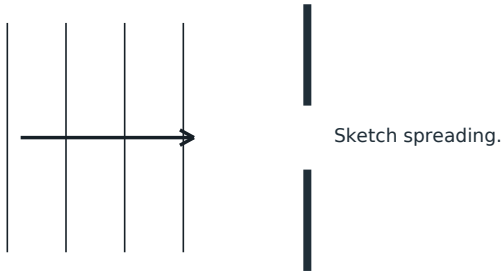
Sketch wavefronts after a narrow opening about one wavelength wide. Compare with a much wider opening.



Marzano check - circle: ① ② ③

04 Apply the physics

A 340 Hz sound travels at 340 m/s toward a doorway 0.8 m wide. Calculate its wavelength and compare wavelength/opening ratios for this sound and visible light of wavelength 5.0×10^{-7} m. Use wavefront sketches to explain why hearing around a doorway differs from seeing around it.



Marzano check - circle: ① ② ③

05 Explain your reasoning

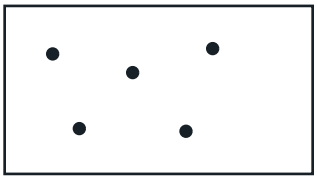
A student calls all wave bending “refraction.” Explain how diffraction can happen without crossing into a different-speed medium.

Marzano check - circle: 1 2 3

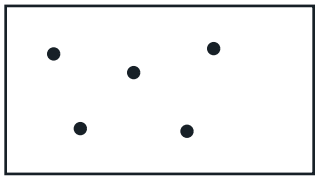
06 Review: connect an earlier idea

For a fixed ideal-gas particle mass, average translational kinetic energy becomes 16 times as large. Compare rms speed.
Use particle arrows and labels to show the comparison.

Sample A



Sample B

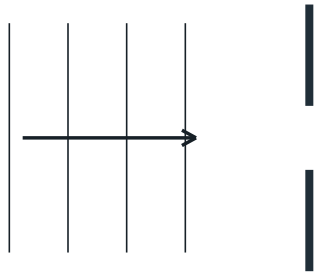


Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: 1 2 3

07 Exit check

At fixed wave speed, frequency doubles. Explain how λ and typical diffraction through a fixed opening change.

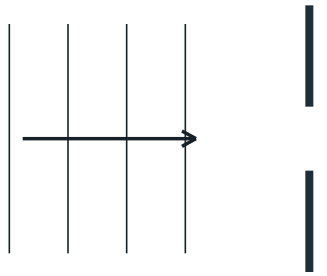


Sketch spreading.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Sound speed is 340 m/s. Compare wavelengths at 170 and 1700 Hz with a 1 m doorway. Explain which should diffract more noticeably.



Sketch spreading.

Marzano check - circle: 1 2 3